

SHARANYA DABAS

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EDUCATION

Cornell University

Ithaca, NY

Bachelor of Science in Computer Science | **GPA:** 3.8

August 2022 – December 2025

Relevant Coursework: Machine Learning, Algorithms, Computer Graphics, Database Systems, Embedded Systems, Computer Networks, Operating Systems, Backend Development, Object-Oriented Programming & Data Structures

EXPERIENCE

Software Development Intern

Boston, MA

State Street Markets | Currency Management

June 2025 – August 2025

- Transformed 100,000+ rows of raw financial data into time-series formats using Pandas to optimize RAG ingestion pipeline
- Developed React + Flask dashboard to help product managers visualize key fund events such as liquidations and mergers
- Wrote SQL queries to track the progress and successful completion of 6 workflow items related to these events
- Created and maintained Oracle SQL schema to model, validate, and store ongoing and historical dashboard event data
- Coordinated weekly calls with the offshore State Street RAG backend team to align data integration methods

Software Developer

Ithaca, NY

Cornell Cup Robotics

January 2024 – May 2025

- Built overhead vision system using April Tags to track location and orientation relative to virtual and physical objects
- Collaborated with 30+ students across 4 sub-teams, participating in weekly scrums and regular milestone demos
- Optimized OpenCV capture pipeline, doubling video frame rate from 30 to 60 fps to support smoother real-time control
- Developed React dashboard with Flask backend for one-click pairing, live location graphing, and movement scripting

PROJECTS

Exercise Search Engine | Python, JavaScript, HTML, CSS

- Lead 5-person team in designing full-stack website for querying exercises and returning similar alternatives
- Created Flask API with SVD text mining and cosine similarity to measure similarity between exercise descriptions
- Wrote a Selenium-based web scraper to compile a JSON dataset from multiple different sources
- Containerized the application via Docker and deployed to AWS ECS using Fargate launch type

Verlet Particle Simulation | TypeScript, Three.js

- Built a Verlet particle simulation using the MVC pattern to separate physics logic, rendering, and user interaction
- Designed a scene graph hierarchy to ensure intuitive motion and interaction between parent, children, and scene
- Implemented rigid-body behavior by linking particles through distance constraints, enabling rotation and collision

Workout Data Visualization App | ASP.NET, React, C#, Javascript, PostgreSQL

- Built fitness tracker web-app allowing users to define exercises, log lifts, and create workout programs
- Developed a React UI with MUI components connected to an ASP.NET Core Web API using Entity Framework Core
- Implemented Identity Framework authentication with SendGrid email integration for secure user management
- Utilized interactive Recharts graphs and a live SVG muscle map that highlights each session's targeted muscle groups
- Designed and optimized a PostgreSQL database on AWS RDS to ensure scalable and reliable data storage

Accelerometer-Driven Game Controller | C, Python

- Developed Doodle Jump on the FRDM-KL46Z, using accelerometer tilt and button interrupts for player input
- Wrote C firmware with PIT-driven LCD score counter, all synced to gameplay via prefixed serial messages
- Designed physics engine that converts raw tilt into vector-based velocity for snappy, believable movement
- Streamed input over UART from the backend to a Pygame front-end for smooth, real-time control

TECHNICAL SKILLS

Languages: Python, C#, Java, C, OCaml, TypeScript, JavaScript, HTML, CSS, SQL

Developer Tools: AWS, PostgreSQL, SQLite, Git, Postman

Frameworks: ASP.NET, React, Entity Framework, Identity Framework, Flask, Docker

Libraries: Pandas, NumPy, OpenCV, Selenium, Matplotlib, Scikit-learn